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COM3502/4502/6502
SPEECH PROCESSING

Lecture 16
Phonology and Prosody

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graph TD; P["PHONOLOGY  
(sound systems)"] --> PH["PHONETICS"]; P --> PM["PHONEMICS"]; PH --> A["auditory ✓"]; PH --> AC["acoustic ✓"]; PH --> AR["articulatory ✓"]; PM --> C["contrast"]; PM --> D["distribution"]; PM --> F["function"];
```

PHONOLOGY
(sound systems)

PHONETICS

auditory ✓
acoustic ✓
articulatory ✓

PHONEMICS

contrast
distribution
function

'physiophonic'
(language-independent)

'psychophonic'
(language-dependent)

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Phonemic Contrast

- The phonetic description of speech is language-independent
- In the previous lectures, I said that only a certain number of phones occur in any given language
... this is not strictly correct
- In principle, all phones can occur in all languages
- However, early phoneticians realised that acoustically distinct speech sounds were only perceived as different by native listeners if they signalled the difference between one word and another in that language
... the sounds are 'contrastive' (*i.e. distinguish one word from another*)
- The **contrastive** phones in a language are called '**phonemes**'

Phonemic Contrast

- English contains the words ...
 - “*fussy*” [fʌsɪ]
 - “*fuzzy*” [fʌzɪ]
- This is known as a '**minimal pair**'
- The existence of a minimal pair is evidence of a 'phonemic contrast'
- Therefore the [s] and [z] sounds are '**phonemes**' in English

Minimal Pairs

- In order to prove that a phonetic distinction is contrastive in a particular language, it is necessary to find a pair of words that differ in *only one* segment
- The differing segment must be in the *same position* in each word
- The two words must mean different things (*i.e. not alternative pronunciations of the same word*)

Examples of Minimal Pairs



- English
 - “light” [laɪt] and “right” [raɪt]
 - “tack” [tæk] and “tax” [tæks]



- French
 - “choux” [ʃy] and “joue” [ʒy]



- Italian
 - “fate” [fato] and “done” [fat:o]



- Spanish
 - “dog” [pero] and “but” [pero]

Finding Minimal Pairs

dot	dock	dog	dots	debt	doll	not	All min. pairs
can	cap	cone	cane	cape	fan	man	2 segments different
hat	that	heart	had	hate	at	hands	3 segments different
green	jean	groan	gain	grease	glean	grain	2 segments different
shoe	show	shoot	shrew	two	blue	who	2 segments different

Highlighted words do not form a minimal pair

The ‘Phoneme’



Daniel Jones
(1881-1967)

- Daniel Jones defined the ‘**phoneme**’ as ...
“a family of uttered sounds (segmental elements of speech) in a particular language which count for practical purposes as if they were one and the same”
- The phonemic inventory of a language is found by exploring all of the possible minimal pairs
- Phonemes are written using IPA symbols
- A ‘**phonemic transcription**’ is an idealised representation of an utterance (whereas a ‘phonetic transcription’ represents the actual sounds used)
- A phonemic transcription is distinguished from a phonetic transcription by using slashes
 - e.g. /fʌsɪ/ and /fʌzɪ/

The 'Phoneme'

- Phonemes might seem to be an abstract concept invented by phoneticians
- However, as an individual acquires their own language, they find it increasingly difficult to both hear and produce non-phonemic differences, e.g.
 - ...
 - Japanese doesn't have a phonemic distinction between [ɹ] and [ɹ̥] (both are 'allophones' of /ɹ/), so adult Japanese listeners can't hear the difference!
 - Arabic has several phonemes that all sound like [h] to a native English listener
- Phonemes are also of huge significance in automatic speech recognition and synthesis (*but are often misunderstood by technologists*)

Moore, R. K., & Skidmore, L. (2019). On the use/misuse of the term 'phoneme'. In *INTERSPEECH-2019*. Graz, Austria.

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Allophones

- Allophones are the set of different phones that make up a given phoneme
 - Japanese: [ɹ] and [ɹ̥] are allophones of /ɹ/
 - English: [p] and [pʰ] are allophones of /p/
 - Spanish: [d] and [ð] are allophones of /d/
 - US English: [tʰ] and [ɾ] are allophones of /t/
- Given a particular language, the distribution of allophones is not random, but is based on the way in which different 'phonetic features' are used to achieve phonemic contrast
 - in English (*and many other languages*) voicing is contrastive for consonants
 - a language in which voicing is not contrastive for consonants allows 'free variation' between voice and voiceless sounds, e.g. in Dyirbal ...
"stone" → [diban], [tiban], [tipan], [dipan]
- Velarisation is *not* contrastive in English, hence [ɹ̠] (as in *"wall"*) and [ɹ] (as in *"light"*) are both allophones of /ɹ/

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Phonological Processes

Phonemic
'intention'

Phonetic
'realisation'

- **'Assimilation'** (*feature spreading*)
 - “can be”
/kæn bi:/ → [kæmbi:] (*place assimilation*)
- **'Elision'** (*deletion*)
 - “I ... do ... not ... know” → “I do not know” →
“I don't know” → “I dunno” → [ɒdnəʊ] → [ɔ̃ɔ̃ɔ̃]
 - “Wednesday” → /wednɜːzdeɪ/ → [wenzdɪ]
- **'Epenthesis'** (*insertion*)
 - “vanilla ice cream”
/vənɪlə aɪskriːm/ → [vənɪləɪaɪskriːm]
- **'Reduction'** (*neutralised vowel quality*)
 - “and” → /ænd/ → [ənd]

Phonological Processes

Phonemic (*idealised citation/dictionary*) form → Phonetic realisation

“fish and chips”

/fɪʃ ənd tʃɪps/ → [fɪʃəndtʃɪps] → [fɪnɪtʃɪps] → [fɪnɪtʃɪps]

“temporary”

/tempərəɪ/ → [tempəɪ] → [temp.ɪ]

“bread and butter”

/bred ənd bʌtə/ → [brednɪbʌtə] → [brebmɪbʌtə] → [brebmʌʔə]

“actually”

/æktʃʊəli:/ → [æktʃʊəli:] → [æktʃʊli:] →
[æktʃəli:] → [ækfəli:] → [ætʃəli:] → [ætʃli:] → [æʃli:]

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Phonological Processes

“Do you actually know any solicitors?”

Phonemic transcription

Orthographic transcription

elision

/du: ju: æktʃʊəli nəʊ enɪ səlɪsɪtɜ:z/

Phonetic transcription

assimilation

epenthesis

reduction

[dʒæʃlɪnəʊwenɪslɪstəz]

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Phonological Processes

Some examples of ...

- ‘**assimilation**’ (feature spreading),
- ‘**elision**’ (deletion),
- ‘**epenthesis**’ (insertion)

elision & assimilation

elision

assimilation & epenthesis

reduction & elision

epenthesis, elision & reduction

– “sandwich” /sændwɪtʃ/ → [sæmwɪtʃ]

– “text processing” /tekst prəʊsesɪŋ/ → [tekspɹəʊsesɪŋ]

– “something” /sʌmθɪŋ/ → [sʌmpfɪŋ]

– “cup of tea” /kʌp ɒv ti:/ → [kʌpəti:]

– “law and order” /lə: ænd ɔ:dɜ:/ → [lə:ɹənɔ:də]

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Prosody

'Prosody' is one way in which speakers are able to signal the important information in what they are saying

It allows different words to be ***emphasised*** and is critical to communicating unambiguous meanings

It's a bit like the 'punctuation' in text, but prosody carries ***much*** more information!

Prosody

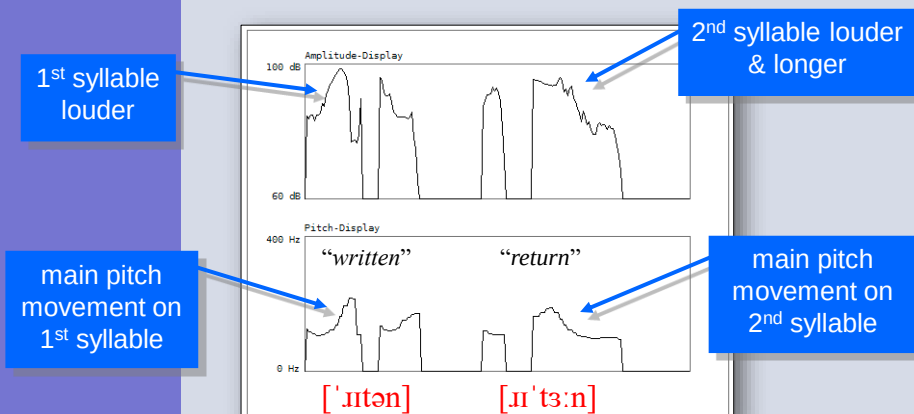
- Prosodic features span ...
 - several speech segments
 - several syllables
 - whole utterances
- Such '**suprasegmental**' behaviour includes ...
 - lexical stress
 - lexical tone
 - rhythmic stress
 - intonation

All carried by
pitch, timing
and loudness

Lexical Stress

- Lexical stress refers to the '**prominence**' of syllables in words
- A stressed syllable is usually **louder** and/or **higher pitch** and/or **longer** than its neighbours, e.g. "billow" vs. "below"
- The location of the stress can be marked using a '**diacritic**', e.g. ['bɪləʊ] vs. [bɪ'ləʊ]

Lexical Stress



Examples of Lexical Stress

“ <u>m</u> ystery”	→ ['mɪstəri:]
“ka <u>n</u> garoo”	→ [kæŋgə 'ru:]
“ <u>f</u> antasy”	→ ['fæntəsi:]
“fa <u>n</u> tastic”	→ [fæn 'tæstɪk]
“e <u>l</u> ephant”	→ ['eləfənt]
“u <u>n</u> derstand”	→ [ʌndə 'stænd]
“i <u>n</u> ten <u>t</u> ion”	→ [ɪn 'tenʃən]
“a <u>f</u> ter <u>n</u> oon”	→ [aftə 'nu:n]
“ <u>c</u> rocodile”	→ ['krɒkədɪəl]
“h <u>o</u> ri <u>z</u> on”	→ [hə 'raɪzən]

Stress Patterns

- The location of lexical stress can vary as a function of different ‘**suffixes**’
 - “photograph” → ['fəʊtəɡrɑf]
 - “photography” → [fə 'tɒɡrəfi]
 - “photographic” → [fəʊtə 'ɡræfɪk]
- The shift of lexical stress away from a syllable often causes a weakening of the vowel
 - vowel quality → [ə]
 - known as vowel ‘**centralisation**’ or ‘**reduction**’
- Lexical stress can be contrastive
 - “billow” → ['bɪləʊ]
 - “below” → [bɪ 'ləʊ]
 - “import” (verb) → ['ɪmpɔ:t]
 - “import” (noun) → [ɪm 'pɔ:t]

Lexical Tone

- About half of the world's languages use the pitch pattern (called '**lexical tone**') to distinguish between one word and another
- Modern Standard Chinese has *four* lexical tones
high level, high rising, low falling-rising, high falling
 [wo mei jou bǐ] → "I haven't got a pen"
 [wo mei jou bī] → "I haven't got a nose"
- Languages with moving pitch patterns (such as Modern Standard Chinese) are called '**contour tone**' languages
- Languages with entirely level tones (such as Yoruba) are called '**register tone**' languages

Lexical Tone

施氏食獅史

石室詩士施氏，嗜獅，誓食十獅。

氏時時適市視獅。

十時，適十獅適市。

是時，適施氏適市。

氏視是十獅，恃矢勢，使是十獅逝世。

氏拾是十獅屍，適石室。

石室濕，氏使侍拭石室。

石室拭，氏始試食是十獅。

食時，始識是十獅，實十石獅屍。

試釋是事。

Lexical Tone

Lion-Eating Poet in the Stone Den

*In a stone den was a poet Shi, who was a lion addict, and had resolved to eat ten.
He often went to the market to look for lions.
At ten o'clock, ten lions had just arrived at the market.
At that time, Shi had just arrived at the market.
He saw those ten lions, and using his trusty arrows, caused the ten lions to die.
He brought the corpses of the ten lions to the stone den.
The stone den was damp. He asked his servants to wipe it.
After the stone den was wiped, he tried to eat those ten lions.
When he ate, he realized that these ten lions were in fact ten stone lion corpses.
Try to explain this matter.*



Lexical Tone

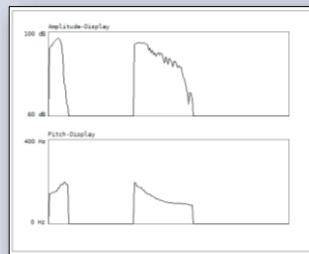
Shī Shì shí shī shǐ

*Shíshì shǐshì Shī Shì, shì shī, shì shí shí shī.
Shì shíshí shì shì shì shī.
Shí shí, shì shí shī shì shì.
Shì shí, shì Shī Shì shì shì.
Shì shì shì shí shī, shì shī shì, shǐ shì shí shī shìshì.
Shì shí shì shí shī shī, shì shíshì.
Shíshì shī, Shì shǐ shì shì shíshì.
Shíshì shì, Shì shǐ shì shí shì shí shī.
Shí shí, shǐ shí shì shí shī, shí shí shí shī shī.
Shì shì shì shì.*

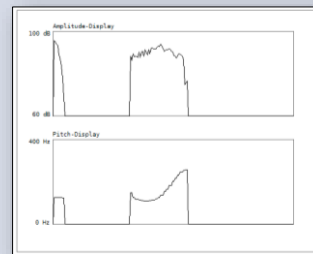
http://en.wikipedia.org/wiki/Lion-Eating_Poet_in_the_Stone_Den

Intonation

Pitch variation that *doesn't* affect the meaning of the words, but *does* affect the meaning of an utterance is known as '**intonation**'

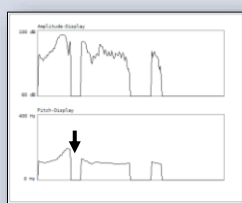


"It's cold!"

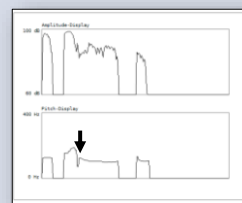


"It's cold?"

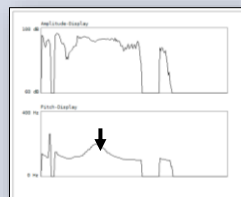
Intonational Accent



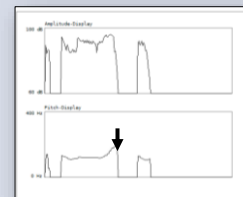
"This is my laptop"



"This is my laptop"



"This is my laptop"



"This is my laptop"

THE INTERNATIONAL PHONETIC ALPHABET (revised to 2015)										
CONSONANTS (PLOSIVES)										© 2015 IPA
	Bilabial	Dental	Alveolar	Postalveolar	Fricative	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p		t		k	c	g	q	ʕ	ʔ
Aspirated	pʰ		tʰ		kʰ	cʰ	gʰ	qʰ		
Unaspirated	p̚		t̚		k̚	c̚	g̚	q̚		
Explosive		v ¹			f					
Labiodental	ɸ ²	β ³	θ ⁴	ð ⁵	ʃ ⁶	ʒ ⁷	s ⁸	z ⁹	x ¹⁰	h ¹¹
Dental			t̪							
Alveolar				ɹ						
Postalveolar					ʈ					
Fricative		v			f					
Labiodental						ç				
Alveolar							x			
Postalveolar								χ		
Uvular								ʁ		
Pharyngeal									ħ	
Glottal										ʕ

¹English in the right of each row is nasal, to the left are oral.
²English does not use bilabial fricative.
³English does not use bilabial plosive.
⁴English does not use dental fricative.
⁵English does not use dental plosive.
⁶English does not use postalveolar fricative.
⁷English does not use postalveolar plosive.
⁸English does not use alveolar fricative.
⁹English does not use alveolar plosive.
¹⁰English does not use uvular fricative.
¹¹English does not use uvular plosive.

SUPRASEGMENTALS		TONES & WORD ACCENTS			
		LEVEL		CONTOUR	
Primary stress	founə'tɪʃən	é	Extra high	ě	Rising
Secondary stress		or	High	or	Falling
Long	e:	é	High	ê	Falling
Half-long	e'	ē	Mid	ě	High rising
Extra-short	ě	è	Low	ě	Low rising
Syllable break	ti.ækt	ẽ	Extra low	ẽ	Rising-falling etc
Minor (foot) group		↓	Downstep	↗	Global rise
Major (intonation) group		↑	Upstep	↘	Global fall
Linking (absence of a break)					

This slide set has covered ...

- Phonetic contrast
- Minimal pairs
- The 'phoneme'
- Allophones
- Assimilation, elision, epenthesis
- Suprasegmental behaviour
- Lexical stress
- Lexical tone
- Rhythmic stress
- Intonation

Any Questions ?



Next time ...

Adaptive Filters